

Vector functions. Divergence and curl. Vector identities.

[SUBMIT ANSWERS TO ALL QUESTIONS]

For a vector function (or *vector field*) $\mathbf{A}$, the divergence is the scalar $\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$.

The curl of $\mathbf{A}$ is the vector $\nabla \times \mathbf{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \mathbf{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \mathbf{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \mathbf{k}$.

Vector identities:

$$\nabla(fg) = \nabla f g + f \nabla g, \quad (1)$$

$$\nabla \cdot (f\mathbf{A}) = \nabla f \cdot \mathbf{A} + f \nabla \cdot \mathbf{A}, \quad (2)$$

$$\nabla \times (f\mathbf{A}) = \nabla f \times \mathbf{A} + f \nabla \times \mathbf{A}, \quad (3)$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}), \quad (4)$$

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla) \mathbf{A} - \mathbf{B} (\nabla \cdot \mathbf{A}) + \mathbf{A} (\nabla \cdot \mathbf{B}) - (\mathbf{A} \cdot \nabla) \mathbf{B}. \quad (5)$$

Second-order derivatives:

$$\nabla \cdot \nabla f \equiv \nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}, \quad (6)$$

$$\nabla \times \nabla f = 0, \quad (7)$$

$$\nabla \cdot \nabla \times \mathbf{A} = 0, \quad (8)$$

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}. \quad (9)$$

Vector field $\mathbf{A}$ is potential (*irrotational*) if $\nabla \times \mathbf{A} = 0$; in this case $\mathbf{A} = \nabla U$ (U is the potential).

Vector field $\mathbf{B}$ is *solenoidal* if $\nabla \cdot \mathbf{B} = 0$; in this case $\mathbf{B} = \nabla \times \mathbf{A}$ ($\mathbf{A}$ is the vector potential).

1. Prove the identities involving $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, $r = \sqrt{x^2 + y^2 + z^2}$ and constant vector $\mathbf{a}$:

$$(a) \nabla r = \mathbf{r}/r, \quad (b) \nabla \cdot \mathbf{r} = 3, \quad (c) \nabla \times \mathbf{r} = 0, \quad (d) \nabla(\mathbf{a} \cdot \mathbf{r}) = \mathbf{a}.$$

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2. Given that $f = x^2y - 2z^2 + y^3$ and $\mathbf{A} = 2x\mathbf{i} - 4xy\mathbf{j} + (z + 2x)\mathbf{k}$, find

$$(a) \nabla f \quad \text{and} \quad \nabla^2 f; \quad (b) \nabla \cdot \mathbf{A} \quad \text{and} \quad \nabla \times \mathbf{A}.$$

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3. Show that $\mathbf{A} = (2xy + 3)\mathbf{i} + (x^2 - 4z)\mathbf{j} - 4y\mathbf{k}$ is irrotational and find the potential U . After finding U , verify that $\nabla U = \mathbf{A}$.

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4. Show that the vector field $\mathbf{B} = f(r)\mathbf{r}$ is solenoidal only if $f(r) = \frac{k}{r^3}$ (k is a constant).

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[Hint: find $\nabla \cdot \mathbf{B}$ using Eq. (2), set it to zero and solve the differential equation for f .]

5. Show that if the vector field $\mathbf{A} = (xyz)^m(x^n\mathbf{i} + y^n\mathbf{j} + z^n\mathbf{k})$ is irrotational (i.e., $\nabla \times \mathbf{A} = 0$) then either $m = 0$ or $n = -1$.

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